INCH-POUND
MIL-PRF-19978J
25 March 2003
SUPERSEDING
MIL-PRF-19978H
25 February 2002

PERFORMANCE SPECIFICATION

# CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC (HERMETICALLY SEALED IN METAL, CERAMIC OR GLASS CASES), ESTABLISHED AND NON-ESTABLISHED RELIABILITY GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the general requirements for established reliability (ER) and non-established reliability (non-ER), direct current (dc), plastic or paper-plastic dielectric, fixed capacitors, hermetically sealed in metal or ceramic or glass cases. Capacitors meeting the established reliability requirements specified herein have failure rate (FR) levels ranging from 1.0 percent to 0.001 percent per 1,000 hours (see 1.2.1.9). These FRLs are established at a 90-percent confidence level and maintained at a 10-percent producer's risk and are based on life tests performed at maximum rated voltage and maximum rated temperature. An acceleration factor of 5:1 has been used to relate the life test data obtained at 140 percent of rated dc voltage at the applicable high test temperature to the rated voltage at the applicable high test temperature. This specification also covers removable mounting retainers for use with applicable capacitors (see 3.1). A part per million (ppm) quality system is used for documenting and reporting the average outgoing quality of ER capacitors supplied to this specification. Statistical process control (SPC) techniques are required in the manufacturing process to minimize variation in production of ER capacitors supplied to the requirements of this specification.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). The PIN is in the following form and as specified (see 3.1).

ER

| CQR09 | A | 1 | M | C | 152 | K | 1 | M |
|---|---|---|---|---|---|---|---|---|
| ER style (1.2.1.1) | Terminal (1.2.1.2) | Circuit (1.2.1.3) | Charac-teristic (1.2.1.4) | Voltage (1.2.1.5) | Capacitance (1.2.1.6) | Capacitance tolerance (1.2.1.7) | Vibration grade (1.2.1.8) | Product level designator (1.2.1.9) |

NON-ER

| CQ09 | A | 1 | M | C | 152 | K | 1 |
|---|---|---|---|---|---|---|---|
| Style (1.2.1.1) | Terminal (1.2.1.2) | Circuit (1.2.1.3) | Charac-teristic (1.2.1.4) | Voltage (1.2.1.5) | Capacitance (1.2.1.6) | Capacitance tolerance (1.2.1.7) | Vibration grade (1.2.1.8) |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC/VAT, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.1 Style The style is identified by either the three-letter symbol "CQR" or the two-letter symbol "CQ" followed by a two-digit number. The letters identify plastic (or paper-plastic) dielectric, fixed capacitor, hermetically sealed in metal, ceramic or glass cases. The symbol "CQR" identifies established reliability (ER) capacitors; the symbol "CQ" Identifies capacitors for which no specific reliability requirements are specified (non-ER). The first digit following the letter symbols identifies the general shape of the case, and the second digit identifies specific details other than case size. Each style designation may Include a family of case sizes.

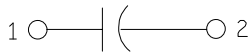
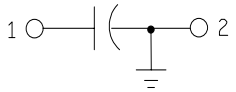
1.2.1.2 Terminal. The terminal is identified by a single letter in accordance with table I.

TABLE I. Terminal.

| Symbol | Type of terminal |
|---|---|
| A | Axial wire lead |
| B | Solder lug (nonremovable) |
| C | Threaded stud and nuts |
| D and H | Pillar insulator for use at altitudes up to 7,500 feet (22.8 inches of mercury) |
| E | Pillar insulator for use at altitudes up to 50,000 feet (3.4 inches of mercury) |

1.2.1.3 Circuit. The circuit diagram and the number of terminals are identified by a single digit in accordance with table II.

TABLE II. Circuit diagram and number of terminals.

| Symbol | Circuit diagram | Number of terminals |
|---|---|---|
| 1 |   | 2 |
| 3 |   | 2 |

1.2.1.4 Characteristic. The characteristic is identified by a single letter in accordance with table III.

* TABLE III. Characteristic.

| Characteristic | Values of characteristics | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | E | F and G | K 2/ | M | P | Q | T | V |
| High ambient test temperature, degrees Celsius ±3°C 1/ | 85 | 85 | 125 | 85 | 65 | 125 | 170 | 125 |
| Low ambient test temperature, degrees Celsius +0°C, -5°C | -65 | -55 | -65 | -65 | -65 | -55 | -65 | -55 |
| Life-test dc voltage, percent of the dc voltage rating (see 4.7.21): | | | | | | | | |
| Watt-second group (see 6.5.3): | | | | | | | | |
| I (0.5 watt-second and less) | 140 | 140 | 140 | 140 | 140 | 150 | 140 | 150 |
| II (0.5+ to 5 watt-seconds) | 140 | 130 | 140 3/ | - - - | - - - | - - - | - - - | - - - |
| III (5+ to 50 watt-seconds) | 140 | 110 | 140 | - - - | - - - | - - - | - - - | - - - |
| IV (greater than 50 watt-seconds) | 140 | 90 | 140 | - - - | - - - | - - - | - - - | - - - |
| Flashpoint of impregnant of filling compound, degrees Celsius | 142 | 135 | 142 | 142 | 142 | 142 | 217 | 142 |

1/ For characteristic K, voltage derating may be necessary at the high ambient test temperature (see 3.1).
2/ For tubular units of characteristic K rated at 1,000 volts dc, life test voltage is 1,200 volts.
3/ For tubular units of characteristic K in watt-seconds group II, use 130 percent of the dc voltage at 40°C for the life-test dc voltage.



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1.2.1.5 Voltage. The dc voltage rating for continuous operation at the high ambient test temperature specified in table III (except for characteristic K which is for 85°C operation), is identified by a single letter in accordance with table IV.

TABLE IV. DC voltage rating.

| Symbol | DC voltage rating 1/ (Volts) | Symbol | DC voltage rating 1/ (Volts) |
|---|---|---|---|
| Z | 30 | K | 2,500 |
| A | 50 | L | 3,000 |
| B | 100 | M | 4,000 |
| C | 200 | N | 5,000 |
| D | 300 | P | 6,000 |
| E | 400 | R | 7,500 |
| F | 600 | S | 10,000 |
| G | 1,000 | T | 12,500 |
| H | 1,500 | U | 15,000 |
| J | 2,000 | | |

1/ For characteristic K, voltage derating may be necessary at the high ambient test temperature (see 3.1).

1.2.1.6 Capacitance. The nominal capacitance value expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

1.2.1.7 Capacitance tolerance. The capacitance tolerance in percent is identified by a single letter in accordance with table V.

TABLE V. Capacitance tolerance.

| Symbol | Capacitance tolerance percent (±) |
|---|---|
| F | 1 |
| G | 2 |
| J | 5 |
| K | 10 |

1.2.1.8 Vibration grade. The vibration grade is identified by a single digit in accordance with table VI.

TABLE VI. Vibration grade.

| Symbol | Frequency range (Hz) | Acceleration (G) |
|---|---|---|
| 1 | 10 to 55 inclusive | - - - |
| 3 | 10 to 2,000 inclusive | 15 |

1.2.1.9 Product level designator. The product level designation in percent per 1,000 hours is identified by a single letter in accordance with table VII, and is based on rated voltage at the high test temperature, as applicable.

TABLE VII. Product level designator.

| Symbol | Product level (percent per 1,000 hours) |
|---|---|
| M | 1.0 |
| P | 0.1 |
| R | 0.01 |
| S | 0.001 |



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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. standards. and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

(See Supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

| | | |
|---|---|---|
| FED-STD-H28 | - | Screw Thread Standards for Federal Services. |

DEPARTMENT OF DEFENSE

| | | |
|---|---|---|
| MIL-STD-202 | - | Test Method Standard Electronic and Electrical Component Parts. |
| MIL-STD-690 | - | Failure Rate Sampling Plans and Procedures. |
| MIL-STD-790 | - | Standard Practice for Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts Specifications. |
| MIL-STD-810 | - | Environmental Engineering Considerations and Consideration Laboratory Tests. |
| MIL-STD-1285 | - | Marking of Electrical and Electronic Parts. |

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

| | | |
|---|---|---|
| ANSI/NCSL Z540-1 | - | General Requirements for Calibration Laboratories and Measuring and Test Equipment. (DoD adopted). |

(Application for copies should be addressed to the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036-0350.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | | |
|---|---|---|
| ASTM D92 | - | Standard Test Method for Flash and Fire Points by Cleveland Open Cup. (DoD adopted). |

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)



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ELECTRONIC INDUSTRIES ALLIANCE (EIA)

| | | |
|---|---|---|
| EIA-554-1 | - | Assessment of Average Outgoing Quality Levels in Parts Per Million (ppm). (DoD adopted). |
| EIA-557 | - | Statistical Process Control Systems. (DoD adopted). |

(Application for copies should be addressed to the Electronic Industries Alliance (EIA), 2500 Wilson Boulevard, Arlington, VA 22201-3834.)

INTERNATIONAL ORGANIZATION FOR STANDARDS (ISO)

| | | |
|---|---|---|
| ISO 10012-1 | - | Quality Assurance Requirements for Measuring Equipment, Part 1: Meteorological Confirmation System for Measuring Equipment. |

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

* 3.2 Qualification. Capacitors furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award. In addition, the manufacturer shall obtain certification from the qualifying activity that the QPL system requirements of 3.3 and 4.2 have been met and are being maintained for the ER styles. Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL.

3.3 Qualified Products List (QPL) system. The manufacturer shall establish and maintain a QPL system for parts covered by this specification. Requirements for this system are specified in MIL-STD-690 and MIL-STD-790. In addition, the manufacturer shall establish a SPC and PPM system which meets the requirements of 3.3.1 and 3.3.2, respectively.

3.3.1 SPC system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish a SPC system which meets the requirements of EIA-557. Typical manufacturing processes for application of a SPC include pre-assembly, assembly, encapsulation, and packaging.

3.3.2 PPM system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish a PPM system for assessing the average outgoing quality of lots in accordance with EIA-554-1. Data exclusion, in accordance with EIA-554-1, may be used with approval of the qualifying activity. The PPM system shall identify the PPM rate at the end of each month and shall be based on a six month moving average.

3.4 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.4.1 Impregnant and filling compounds. Compounds used in the impregnation and filling of capacitors shall be chemically inactive with respect to the capacitor element and the case (see 3.5.1). The compound, either in the state of original application or as a result of having aged, shall have no adverse effect on the performance of the capacitor. For liquid-filled capacitors, the same material shall be used for impregnating as is used for filling (see 6.3).



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3.5 Interface and physical dimension requirements. Capacitors and retainers shall meet the interface and physical dimensions specified (see 3.1).

3.5.1 Case. Each capacitor shall be enclosed in a hermetically-sealed case (see 3.1) which will protect the capacitor element from moisture, impregnant or filling compound leakage, and mechanical damage under all test conditions specified herein.

3.5.2 Sleeving (when applicable, see 3.1). The sleeving material shall not soften, creep, or shrink to a point where any part of the cylindrical portion of the case is left uncovered at any test temperature specified herein. The sleeving shall not obscure the part marking.

3.5.2 Terminals.

3.5.2.1 Case as terminal. When the case is used as a terminal, any protective coating applied to the mounting surfaces shall be such as to provide a direct conducting path for an electric current from the case to the surface on which it is mounted.

3.5.2.2 Solder lugs and solder-lug terminals. Solder lugs and solder lug-terminals may be of any shape, provided dimensional limits are met, and shall be solder coated.

3.5.2.3 Screw terminals. Screw terminals shall be supplied with one nut, one flat washer, and one lockwasher.

3.5.2.4 Wire leads and pins. Leads and pins shall be solder coated.

3.5.2.5 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.7). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is three percent. Tin-lead finishes shall be used only when other plating cannot meet the intended performance requirement.

3.5.3 Threaded parts. All threaded parts shall be as specified (see 3.1) and in accordance with FED-STD-H28.

3.5.3.1 Engagement of threaded parts. All threaded parts shall engage by at least three full threads.

3.5.3.2 Locking of screw-thread assemblies. All screw-thread assemblies shall be rendered resistant to loosening under vibration. Lockwashers shall be provided under all nuts.

3.6 Burn-in (ERs only. when applicable. see 3.1). When capacitors are tested as specified in 4.7.2, there shall be no evidence of damage, arcing, or breakdown.

3.7 Radiographic inspection (ERs only. when applicable. see 3.1). When capacitors are tested as specified in 4.7.3, x-ray examination shall disclose no evidence of improperly made connections, misalignments of seals or eyelets, substandard soldering or structural weakness, or solder particles or slivers attached to one end.

3.8 Thermal shock. When tested as specified in 4.7.4, capacitors and retainers shall withstand the extremes of high and low temperatures without visible damage.

3.9 Seal. When capacitors are tested as specified in 4.7.5, there shall be no continuous visible stream of bubbles or other evidence of leakage.

3.10 Dielectric Withstanding Voltage (DWV).

3.10.1 Capacitors. When capacitors are tested as specified in 4.7.6.1, there shall be no momentary or intermittent arcing or other indication of breakdown, nor shall there be any visible evidence of damage.

3.10.2 Sleeving (when applicable. see 3.1). When capacitors are tested as specified in 4.7.6.2, the insulating sleeve shall withstand the specified potential without breakdown.

3.11 Barometric pressure (reduced) (when applicable. see 3.1), qualification only. When capacitors are tested as specified in 4.7.7, there shall be no momentary or intermittent arcing or other indication of breakdown, nor shall there be any visible evidence of damage.



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3.12 Insulation resistance (IR).

3.12.1 Sleeving (when applicable. see 3.1). When measured as specified in 4.7.8, the insulation resistance shall be not less than 1,000 megohms.

3.12.2 Terminal to terminal. When measured as specified in 4.7.8, the insulation resistance shall be not less than the applicable values specified (see 3.1).

3.12.3 Terminal to case. Unless otherwise specified (see 3.1), when measured as specified in 4.7.8, the insulation resistance between any terminal and case, when the case is not a terminal, shall exceed 50,000 megohms.

3.13 Capacitance. When measured as specified in 4.7.9, the capacitance shall be within the tolerance specified (see 3.1).

3.14 Dissipation factor. When measured as specified in 4.7.10, the dissipation factor shall not exceed the applicable value specified (see 3.1).

3.15 Vibration. When capacitors are tested as specified in 4.7.11, there shall be no intermittent contacts of 0.5 milliseconds (ms) or greater duration or momentary arcing, or other indication of breakdown, nor shall there be any open-or-short circuiting or evidence of mechanical damage. In addition, retainers, as tested to 4.7.11, shall exhibit no evidence of mechanical damage.

3.16 Salt spray (corrosion) (metal surfaces only, see 3.1). When capacitors or retainers are tested as specified in 4.7.12, there shall be no evidence of harmful corrosion and at least 90 percent of any exposed metal surfaces of the capacitor or retainer shall be protected by the finish. Harmful corrosion shall be construed as any type of corrosion which in any way interferes with the mechanical or electrical performance of the capacitor or retainer, as applicable. For capacitors or retainers with painted surfaces, not more than 10 percent of the surfaces shall be affected by flaking, peeling, or blistering of paint. There shall be no evidence of unwrapping of or mechanical damage to the insulating sleeves when applicable. In addition, corrosion of the terminal hardware or mounting surface shall not exceed 10 percent of the surface area. Marking shall remain legible.

3.17 Immersion. When tested as specified in 4.7.13, capacitors or retainers, as applicable, shall meet the following requirements:

a. Capacitors:

Dielectric withstanding voltage:

| | |
|---|---|
| Insulating sleeves (when applicable, see 3.1): | Not less than 4,000 V dc. |
| Terminal to terminal: | As specified in 3.10.1. |
| Terminal to case: | As specified in 3.10.1. |

Insulation resistance at 25°C:

| | |
|---|---|
| Insulating sleeves (when applicable, see 3.1): | Not less than 1,000 megohms. |
| Terminal to terminal: | Not less than 1,000 megohms. Unless otherwise specified, not less than 60 percent of the value specified in 3.12.2. |
| Terminal to case: | Unless otherwise specified, not less than 50 percent of the value specified in 3.12.3. |



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| | |
|---|---|
| Capacitance: | Shall change not more than ±1.0 percent from the value obtained when measured as specified in 4.7.9. |
| Dissipation factor: | Shall not exceed 120 percent of the initial value specified (see 3.1). |

b. Capacitors and retainers:

| | |
|---|---|
| Visual examination: | There shall be no harmful or extensive corrosion of the capacitors or retainers. The marking shall remain legible. There shall be no unwrapping of, or mechanical damage to, the insulating sleeves, when applicable. |

3.18 Solderability (wire leads only. see 3.1). When capacitors are tested as specified in 4.7.14, the dipped portion of the terminals shall conform to the solid-wire termination criteria of method 208 of MIL-STD-202.

3.19 Shock (specified pulse). When capacitors are tested as specified in 4.7.15, there shall be no intermittent contacts of 0.5 ms or greater duration, or arcing or other indications of breakdown, nor shall there be any open or short circuiting or evidence of mechanical damage.

3.20 Terminal strength. When capacitors are tested as specified in 4.7.16, there shall be no mechanical damage to the capacitor or terminals.

3.21 Dielectric absorption (when specified. see 3.1). When measured as specified in 4.7.17, the dielectric absorption shall not exceed the value specified (see 3.1).

3.22 Stability at low end high temperatures. When capacitors are tested as specified in 4.7.18, there shall be no indication of breakdown or arcing, nor shall there be any open or short circuiting or any visible evidence of mechanical damage. The capacitance changes at the specified temperatures shall not exceed the applicable limits specified (see 3.1) from value at 25°C.

3.23 Temperature coefficient (characteristic P only) (See 3.1). When measured as specified in 4.7.19, the temperature coefficient shall not exceed -120 ppm per degree Celsius (ppm/°C) ±50 ppm/°C.

3.24 Life. When tested as specified in 4.7.20, capacitors shall meet the following requirements.

| | |
|---|---|
| a. Insulation resistance at 25°C: | Unless otherwise specified, not less than 60 percent of the value specified in 3.12.1. |
| b. Capacitance: | Shall change not more than the percent specified (see 3.1) of the initial value obtained when measured as specified in 4.7.9. |
| c. Dissipation factor: | Unless otherwise specified, shall not exceed the initial value specified (see 3.1). |
| d. Visual examination: | There shall be no leakage of impregnant or filling compound or deformation of the case either during or after the test. |

3.25 Fungus. The manufacturer shall certify that all external materials are fungus resistant or shall perform the test specified in 4.7.21. When capacitors are tested as specified in 4.7.21, examination shall disclose no evidence of fungus growth on the external surface.

3.26 Resistance to solvents. When capacitors are tested as specified in 4.7.22, there shall be no evidence of mechanical damage and the marking shall remain legible.



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3.27 Resistance to soldering heat (wire-lead capacitors only). When tested as specified in 4.7.23, capacitors shall meet the following requirements:

| | |
|---|---|
| a. Insulation resistance at 25°C: | As specified in 3.12. |
| b. Capacitance: | Change not more than five percent from the Initial value obtained when measured as specified in 4.7.9. |
| c. Dissipation factor: | Shall not exceed initial limit. |

3.28 Moisture resistance. When tested as specified in 4.7.24, capacitors or retainers as applicable, shall meet the following requirements.

a. Capacitors:

| | |
|---|---|
| Dielectric withstanding voltage: | |
| Insulating sleeves (when applicable, see 3.1): | Not less than 4,000 V dc. |
| Terminal to terminal: | As specified in 3.10.1. |
| Terminals to case (circuit diagram 1 only): | As specified in 3.10.1. |
| Insulation resistance at 25°C: | |
| Insulating sleeves (when applicable, see 3.1): | Not less than 1,000 megohms. |
| Terminal to terminal: | Not less than 60 percent of the value specified in 3.12.2. |
| Terminals to case (circuit diagram 1 only): | Not less than 50 percent of the value specified in 3.12.3. |
| Capacitance: | Shall change not more than ±0.5 percent (±2.0 percent for characteristic M) from the value obtained when measured as specified in 4.7.9. |
| Dissipation factor: | Shall not exceed 120 percent of the initial values specified (see 3.1). |

b. Capacitors and retainers:

| | |
|---|---|
| Visual examination: | There shall be no harmful or extensive corrosion of the capacitors or retainers. The marking shall remain legible. There shall be no unwrapping of, or mechanical damage to, the insulating sleeves when applicable. |

3.29 Flashpoint of impregnant or filling compound. Unless otherwise specified (see 3.1), when measured as specified in 4.7.25, the flashpoint of impregnant or filling compound shall be equal to or greater than 142°C.



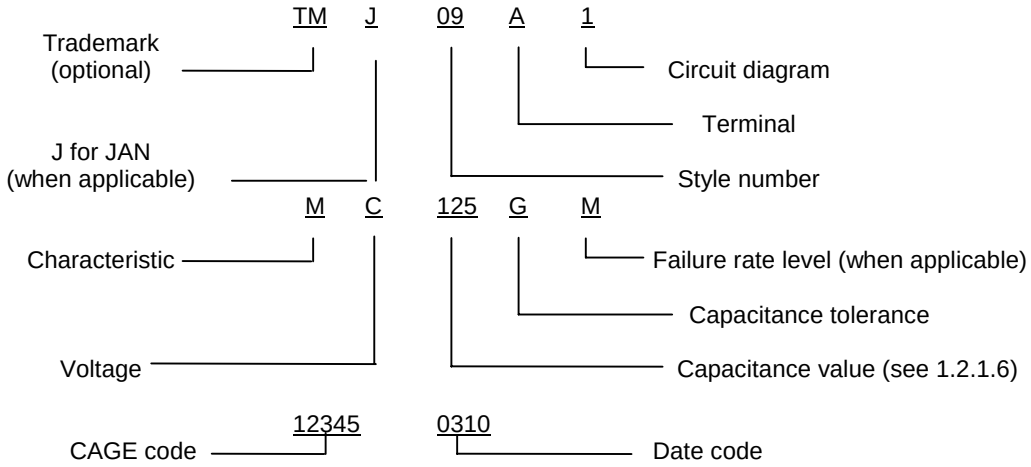
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3.30 Marking.

3.30.1 Capacitors. Marking of capacitors shall conform to method I of MIL-STD-1285 and shall include the PIN, "JAN" brand (ER styles only), trademark (optional), commercial and Government entity (CAGE) code, date code, lot symbol, and when space permits, capacitance (in µF), capacitance tolerance, and rated voltage, and when applicable, failure rate level. Marking shall not be applied to the insulating sleeve. Unless otherwise specified (see 3.1), capacitors shall be marked as shown in the following example:

| | | | |
|---|---|---|---|
| Example: | CQR09A1MC152K1M | - | PIN. |
| | JAN TM 12345 | - | "JAN" brand (ER styles only), trademark (optional) and CAGE code. |
| | 0150A.0011µF | - | Date code, lot symbol, and capacitance. |
| | 10% 1,000 V dc | - | Capacitance tolerance and rated voltage. |

3.30.2 Partial marking. For tubular capacitors with case size of .15 by .406, .105 by .406, and .235 by .406, the following partial marking may be used.




3.30.3 Retainer. Retainers shall be marked with the PIN, trademark (optional), and CAGE code. Marking shall be on the outer surface of the retainers.

3.30.4 JAN and J marking (ER styles only). The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of specifications. Accordingly, items acquired to and meeting all of the criteria specified herein and in applicable specification, shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the PIN except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the PIN. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated specifications, the manufacturer shall remove completely the military part number and the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 1,586,261 for the certification mark "J".

3.30.5 Substitution of failure rate levels (FRL's). A manufacturer may supply to all higher FRL than to which they are qualified. Items of an exponential FRL as shown in table VIII and marked to lower FRL with procuring agency approval, are substitutable for higher FRL, and shall not be remarked unless specified in the contract or purchase order (see 6.2), the lot date codes on the parts are unchanged, and the workmanship criteria is met.



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TABLE VIII. FRL substitutability.

| Parts qualified to FRL | Substitutable for FRL |
|---|---|
| S | M, P, and R |
| R | M and P |
| P | M |

3.30.6 Substitution of capacitance tolerance and voltage. Parts qualified and marked to tighter capacitance tolerance or higher rated voltage, with procuring agency approval, are substitutable for parts marked to looser capacitance tolerance or lower rated voltage, provided all other values, such as case size, characteristic, and leads are the same. The substitutable parts shall not be remarked unless specified in the contract or purchase order (see 6.2), the lot date code on the parts are unchanged, and the workmanship criteria are met.

3.30.7 Non-ER marking. An ER part may be marked and furnished as a non-ER part if produced on the same assembly line, and provided it is subject to and meets all the inspection requirements of the ER part.

3.31 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.32. Workmanship. Capacitors and retainers shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (ER styles: See 4.5; non-ER: See 4.5.1).

c. Conformance inspection (see 4.6).

4.2 QPL system (applicable to ER parts only). The manufacturer shall establish and maintain a QPL system in accordance with 3.3. Evidence of such compliance is a prerequisite for qualification and retention of qualification.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with ANSI/NCSL Z540-1, ISO-10012-1, or equivalent system.

4.3.2 AC measurements. Alternating-current (ac) measurements shall be made at the frequency specified. The magnitude of the ac voltage shall be limited to 1.0 volt root-mean-square (rms). The maximum dc bias voltage shall be 2.2 volts for all ac measurements of capacitors.

4.3.3 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at 25°C ±5°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.4 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the rated voltage. The power source employed for dc leakage current measurements shall be stabilized to at least 100 ppm. No voltage fluctuations shall occur during measurements that would produce a variation in the current measurement as read with any acceptable dc leakage current tester used to test capacitors.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.4) on sample units produced with equipment and procedures normally used in production.



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* 4.4.1 Sample size. The number of capacitors to be submitted for qualification inspection shall be as specified in table IX or the appendix to this specification.

* 4.4.2 Inspection routine. Capacitors shall be subjected to the qualification inspection specified in table IX in the order shown. All sample units shall be divided as specified in table IX for groups I through VIII inclusive, and subjected to the inspection for their particular group; for combined voltage group submissions, each type shall be equally represented in each group.

* 4.4.3 Failures. Failures in excess of those allowed in table IX shall be cause for refusal to grant qualification approval.

4.4.4 Failure rate (FR) qualification (applicable to ER parts only). FR qualification for capacitors shall be in accordance with the general and detailed requirements of MIL-STD-690 and as follows.

a. Procedure I: Qualification at the initial FRL. Level "M" (1.0 percent) of FRSP-90 shall apply. Sample units shall be subjected to the qualification inspection specified in group V, table IX (see 4.4.2). The entire life test sample shall be continued on test to 6,000 hours as specified in 4.7.20.3, upon completion of the 2,000 hour qualification.

* b. Procedure II: Extension of qualification to lower FRLs. To extend qualification to the "R" (0.01 percent) and "S" (0.001 percent) FRLs, data from two or more styles of similar construction (see 4.6.1.1.1) may be combined.

c. Procedure Ill: Maintenance of FRL qualification. Maintenance period B of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.6.1).

4.4.5 Quality level verification (ER styles only). The manufacturer is responsible for establishing a quality system to assess the ppm defect level of lots that are subjected to subgroup 2 tests of the group A inspections. The ppm defect level shall be based on a 6-month moving average.

4.5. Verification of qualification (ER styles only). Every 12 months the manufacturer shall provide verification of qualification to the qualifying activity. Continuation of qualification shall be based on meeting the following requirements:

a. MIL-STD-790 program.

b. The capacitor design has not been modified.

c. Lot rejection for group A inspection does not exceed 5 percent or one lot, whichever is greater.

d. The requirements for group B inspection are met.

e. Verification of FRLs.

f. PPM assessment.

In the event that there is no production of a single style device during a maintenance period and the manufacturer is listed for more than one style on the QPL, the manufacturer shall certify that they retain the capabilities and facilities necessary to produce that product. However, the manufacturer shall still maintain the required number of unit hours in the maintenance period using those styles produced in order to remain qualified to the applicable failure rate levels. In the case where the lowest failure rate for an un-produced style is M, styles need not be manufactured for testing only but the manufacturer must certify that the capability and facilities needed to produce that style are still in place. In the event that units must be built for the purpose of maintaining the required hours, they shall also undergo all required testing prior to being placed on life test. If during three consecutive reporting periods there has been no production of a given style, the manufacturer may be required, at the discretion of the qualifying activity, to submit a newly-produced (not from stock) representative product of that style to testing.



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TABLE IX. Qualification inspection.

| Inspection | Requirement paragraph | Method Paragraph | NON-ER | | ER | | |
|---|---|---|---|---|---|---|---|
| | | | Number of sample units to be inspected | Number of defectives allowed 1/ | Number of sample units to be inspected | Number of defectives allowed 1/ | |
| Group I<br>Visual and mechanical examination (internal):<br>Material and workmanship | 3.1, 3.4, and 3.32 | 4.7.1 | 2 | 0 | 2 | 0 | |
| Group II<br>Visual and mechanical examination (external):<br>Physical dimensions, marking, and workmanship 2/<br>Burn-in (ER only, when applicable) 3/<br>Radiographic inspection (ER only, when applicable) 3/<br>Thermal shock 3/<br>Seal<br>Dielectric withstanding voltage 3/<br>Barometric pressure (reduced) (when applicable) 3/<br>Insulation resistance 3/<br>Capacitance 3/<br>Dissipation factor 3/ | 3.1, 3.5, and 3.30 to 3.32<br>3.6<br>3.7<br>3.8<br>3.9<br>3.10<br>3.11<br>3.12<br>3.13<br>3.14 | 4.7.1<br>4.7.2<br>4.7.3<br>4.7.4<br>4.7.5<br>4.7.6<br>4.7.7<br>4.7.8<br>4.7.9<br>4.7.10 | 43 | 1 | 80 | 0 | |
| Group III<br>Vibration<br>Salt spray (corrosion) (metal case only)<br>Immersion | 3.15<br>3.16<br>3.17 | 4.7.11<br>4.7.12<br>4.7.13 | 6 | 1 | 12 | 1 | 1 |
| Group IV<br>Solderability (wire leads only) | 3.18 | 4.7.14 | 6 | 1 | 6 | 1 | |
| Group V<br>Shock (specified pulse)<br>Terminal strength<br>Moisture resistance | 3.19<br>3.20<br>3.28 | 4.7.15<br>4.7.16<br>4.7.24 | 6 | 1 | 12 | 1 | |
| Group VI<br>Dielectric absorption (when specified)<br>Stability at low and high temperatures 3/<br>Temperature coefficient (characteristic P)<br>Life | 3.21<br>3.22<br>3.23<br>3.24 | 4.7.17<br>4.7.18<br>4.7.19<br>4.7.20 | 24 | 1 | 40 | 1 | |
| Group VII<br>Fungus<br>Resistance to solvents<br>Resistance to soldering heat (wire lead only)<br>Moisture resistance | 3.25<br>3.26<br>3.27<br>3.28 | 4.7.21<br>4.7.22<br>4.7.23<br>4.7.24 | N/A | N/A | 5<br>5 | 10 | 1 |
| Group VIII<br>Flashpoint of impregnant or filling compound | 3.29 | 4.7.25 | 2 4/ | N/A | 2 4/ | N/A | |

1/ A sample unit having one or more defects will be considered as a single defective.
2/ Marking defects are based on visual examination only and will be charged only for illegible, incomplete, or incorrect marking.
3/ Nondestructive tests.
4/ Two samples shall be taken from the .25-pound impregnant or filling compound submitted.



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4.5.1 Verification of qualification (non-ER styles only). Every 12 months, the manufacturer shall provide verification of qualification to the qualifying activity. Continuation of qualification shall be based on meeting the following requirements:

a. Group A inspection.

b. Periodic group B inspection.

In the event that there is no production of a single style device during a reporting period and the manufacturer is listed for more than one style on the QPL, the manufacturer shall certify that they retain the capabilities and facilities necessary to produce that product. If during three consecutive reporting periods there has been no production of a given style, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of that style to testing.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 Inspection and production lot.

* 4.6.1.1.1 Inspection lot. An inspection lot shall consist of all capacitors in one or more styles, produced under essentially the same conditions, and offered for inspection during a single work month. The sample selected from the lot shall be representative of the styles in the lot. ER parts shall be kept separate from non-ER parts. The following styles are considered to be of similar style and may be combined for conformance and FRL maintenance:

| Group | Characteristic | |
|---|---|---|
| 1 | K, M | CQR09, CQR12, CQR13, CQR19, CQR29, CQR32, CQR33, CQR39, CQR42, CQR43. |
| 2 | Q, V | CQR07 and CQR44 (May be maintained by MIL-PRF-83421/6). |
| 3 | E, F, G, or K | CQ72. |
| 4 | | All non-ER styles may be grouped by characteristic. |

4.6.1.1.2 Production lot. A production lot shall consist of all capacitors of the same style, voltage rating, nominal capacitance value, and voltage-temperature characteristic. The manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examination and tests specified in table X or table XIII.



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TABLE X. Group A inspection (ER).

| Inspection | Requirement paragraph | Test method paragraph | Sampling procedure |
|---|---|---|---|
| Subgroup 1 | | | |
| Burn in | 3.6 | 4.7.2 | |
| Radiographic inspection 1/ | 3.7 | 4.7.3 | |
| Thermal shock | 3.8 | 4.7.4 | |
| Seal | 3.9 | 4.7.5 | 100% inspection |
| Dielectric withstanding voltage | 3.10 | 4.7.6 | |
| Insulation resistance | 3.12 | 4.7.8 | |
| Capacitance | 3.13 | 4.7.9 | |
| Dissipation factor | 3.14 | 4.7.10 | |
| Subgroup 2 | | | |
| Visual and mechanical examination (external): | | | |
| Physical dimensions | 3.1, 3.5, and 3.30 to 3.32 inclusive | 4.7.1 | See table XI |
| Marking 2/ | | | |
| Workmanship | | | |
| Subgroup 3 3/ | | | |
| Solderability (wire lead styles only) | 3.18 | 4.7.14 | 5 samples 0 failures |

1/ Radiographic not applicable to FRL M.
2/ Marking defects are based on visual examination only.
3/ The manufacturer may request the deletion of the subgroup 3 solderability test, provided an in-line or process control system for assessing and assuring the solderability of leads can be validated and approved by the qualifying activity. Deletion of the test does not relieve the manufacturer from meeting this test requirement in case of dispute. If the design, material, construction, or processing of the part is changed or if there are any quality problems, the qualifying activity may require resumption of the test.

TABLE XI. Sampling plans for subgroup 2.

| Lot size | Sample size |
|---|---|
| 1 - 13 | 100% |
| 14 – 125 | 13 |
| 126 – 150 | 13 |
| 151 – 280 | 20 |
| 281 – 500 | 29 |
| 501 – 1,200 | 34 |
| 1,201 – 3,200 | 42 |
| 3,201 - 10,000 | 50 |
| 10,001 - 35,000 | 60 |
| 35,001 - 150,000 | 74 |
| 150,001 - 500,000 | 90 |
| 500,001 - Up | 102 |



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TABLE XII. Group A inspection (Non-ER).

| Inspection | Requirement paragraph | Test method paragraph | Sampling procedure |
|---|---|---|---|
| Subgroup 1 | | | |
| Thermal shock | 3.8 | 4.7.4 | |
| Seal | 3.9 | 4.7.5 | |
| Dielectric withstanding voltage | 3.10 | 4.7.6 | See table XIII |
| Insulation resistance | 3.12 | 4.7.8 | |
| Capacitance | 3.13 | 4.7.9 | |
| Dissipation factor | 3.14 | 4.7.10 | |
| Subgroup 2 | | | |
| Visual and mechanical examination (external): | | | |
| Physical dimensions | 3.1, 3.5, and 3.30 to 3.32 inclusive | 4.7.1 | 13 samples 0 failures |
| Marking 1/ | | | |
| Workmanship | | | |
| Subgroup 3 2/ | | | |
| Solderability (wire lead styles only) | 3.18 | 4.7.14 | 5 samples 0 failures |

1/ Marking defects are based on visual examination only.
2/ The manufacturer may request the deletion of the subgroup 3 solderability test, provided an in-line or process control system for assessing and assuring the solderability of leads can be validated and approved by the qualifying activity. Deletion of the test does not relieve the manufacturer from meeting this test requirement in case of dispute. If the design, material, construction, or processing of the part is changed or there are any quality problems, the qualifying activity may require resumption of the test.

TABLE XIII. Sample size for subgroup 1.

| Lot size | Sample size |
|---|---|
| 1 - 125 | 100% |
| 126 - 3,200 | 125 |
| 3,201 - 10,000 | 192 |
| 10,001 - 150,000 | 294 |
| 150,001 - 500,000 | 345 |
| 500,001 - Up | 435 |

4.6.1.2.1 ER styles.

4.6.1.2.1.1 Subgroup 1 tests. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Capacitors failing the tests of subgroup 1 shall be removed from the lot. If, during the 100 percent inspection, screening required that more than 5 percent of the capacitors be discarded, the entire production lot shall be rejected.

4.6.1.2.1.1.1 Manufacturer's production inspection. If the manufacturer performs tests similar to that specified in subgroup 1, table X, as the final step of their production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with.

a. Manufacturer’s production tests are identical to, or more stringent than, that specified for subgroup 1.

b. One hundred percent of the product is subjected to these tests.

c. Failure criteria are identical to, or more stringent than, those specified herein.



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d. The lot rejection criteria are identical to, or more stringent than, that specified herein.

e. Once approved, the manufacturer shall not change the test procedure or criteria without prior notification and concurrence by the qualifying activity.

4.6.1.2.1.2 Subgroup 2 tests.

4.6.1.2.1.2.1 Sampling plans. Subgroup 2 shall be performed on an inspection lot basis. The sampling procedure shall be as specified in table X.

4.6.1.2.1.2.2 Rejected lots. The rejected lot shall be segregated from new lots and those lots that have passed inspection. The rejected lot shall be 100 percent inspected for those quality characteristics found defective in the sample and any defectives found shall be removed from the lot. A new sample of parts shall then be randomly selected in accordance with table X. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.1.3 Subgroup 3 (solderability).

4.6.1.2.1.3.1 Inspection lot. An inspection lot for the purpose of this test shall consist of all capacitors manufactured with the same diameter lead wire and offered for inspection at one time. Each production lot shall be kept separate from every other lot. All samples belonging to a production lot shall be identified to that lot. Means of identification is at the option of the manufacturer.

4.6.1.2.1.3.2 Sampling plan. Five samples shall be selected randomly from each inspection lot. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more failures, the lot shall be rejected.

4.6.1.2.1.3.3 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot.

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.6.1.2.1.3.2. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in 4.6.1.2.1.3.3b.

b. The manufacturer submits the failed lot to a 100 percent solder dip using a approved solder dip process in accordance with the appendix. Following the solder dip, the electrical measurements required in group A, subgroup 1 tests shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. (NOTE: If x-ray and hermetic seal are required in the group A, subgroup 1 tests, these tests shall be repeated.) Five additional samples shall then be selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails this second rework, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

4.6.1.2.1.3.4 Disposition of sample. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract.

4.6.1.2.2 Non-ER styles.

4.6.1.2.2.1 Sampling plan. The inspection lot shall be submitted to the sampling plan for subgroup 1 and subgroup 2 as specified in table XII.

4.6.1.2.2.2 Rejected lots. If an inspection lot for subgroup 1 or subgroup 2 is rejected, the contractor may rework it to correct the defects, or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.6.1.2.2.3 Subgroup 3 (solderability). Five samples shall be selected randomly from each inspection lot. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more failures, the lot shall be rejected.



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4.6.1.2.2.3.1 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot.

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.6.1.2.1.3.2. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in 4.6.1.2.2.3.1b.

b. The manufacturer submits the failed lot to a 100 percent solder dip using a approved solder dip process in accordance with the appendix. Following the solder dip, the electrical measurements required in group A, subgroup 1 tests shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Five additional samples shall then be selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails this second rework, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

4.6.1.2.2.3.2 Disposition of sample. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract.

4.6.2 PPM calculations (non-ER and ER). The manufacturer shall establish and maintain a system for assessing and calculating average outgoing quality of capacitors. A PPM rate combining IR, capacitance, and DF shall be assessed for lots that have passed the group A inspection. The manufacturer's PPM system shall also address rectification procedures for lots failing PPM assessment. Data from the rectification process shall not be used to calculate PPM.

4.6.3 Periodic group B inspection. Periodic group B inspection shall consist of the tests specified in table XIV in the order shown, and shall be performed on sample units selected from lots that have passed group A inspection. Except where the results of this inspection show noncompliance with the applicable requirements (see 4.6.3.4), delivery of products which have passed group A shall not be delayed pending the results of this periodic inspection.

4.6.3.1 Sampling plan.

* 4.6.3.1.1 For all non-ER styles and FRLs. Sample units shall be selected from the first lot and then from every two months production for subgroups 1, 2, 3, 4, and 5. The highest watt-second rating in each style, characteristic, and voltage manufactured during the specified periods shall be represented in at least the approximate ratio of production. A different sample shall be selected for each subgroup.

4.6.3.1.2 For FRLs M, P, R, and S. In addition to the tests specified in 4.6.3.1.1, a minimum ten sample units from each inspection lot (see 4.6.3.1.1) shall be subjected to subgroup 3 of table XIV. Allowable failures shall be as specified in table IV of MIL-STD-690.

4.6.3.2 Failures. If the number of failures exceeds the number allowed in table XIV, the sample shall be considered to have failed.

4.6.3.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.6.3.4 Noncompliance (applicable to both ER and non-ER parts). If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the qualifying activity and the contracting officer or purchaser.



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* TABLE XIV. Group B inspection

| Inspection | Requirement paragraph | Method paragraph | Number of samples | Number of defectives allowed | | | |
|---|---|---|---|---|---|---|---|
| | | | | Non-ER | | ER | |
| Subgroup 1<br>Shock (specified pulse) 1/ 2/<br>Vibration 2/<br>Salt spray (corrosion) (metal cases only) 3/<br>Immersion 3/ | 3.19<br>3.15<br>3.16<br>3.17 | 4.7.15<br>4.7.11<br>4.7.12<br>4.7.13 | 12 | 1 | 1 | 1 | 1 |
| Subgroup 2<br>Terminal strength 3/<br>Moisture resistance 3/<br>Dielectric withstanding voltage (sleeving only) 3/ | 3.20<br>3.28<br>3.10.2 | 4.7.16<br>4.7.24<br>4.7.6.2 | 12 | 1 | 1 | 1 | 1 |
| Subgroup 3<br>Life | 3.24 | 4.7.20.2 | 12 (Non-ER)<br>ER (10 per inspection lot minimum) | 1 | | see 4.6.3.1 | |
| Subgroup 4<br>Insulation resistance (at high ambient Test temperature)<br>Dielectric absorption (when specified)<br>Stability at low and high temperatures<br>Temperature coefficient (characteristic P only) | 3.12<br>3.21<br>3.22<br>3.23 | 4.7.8<br>4.7.17<br>4.7.18<br>4.7.19 | 12 | 1 | | 1 | |
| Subgroup 5<br>Resistance to solvents 3/<br>Resistance to soldering heat (wire-lead styles only) 3/ | 3.26<br>3.27 | 4.7.22<br>4.7.23 | 6 | N/A | | 1 | |

1/ Not applicable to ceramic or glass-cased units
2/ The time period for shock and vibration tests is a minimum of once every 24 months.
3/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, this test, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform this test every three years after the deletion as part of long term design verification. If the design, material, construction, or processing of the part is changed, or if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirements in case of dispute.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. Capacitors and retainers shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4 to 3.5.3 inclusive, 3.30 to 3.30.7 inclusive, and 3.32).

4.7.2 Burn-in (see 3.6) (ERs only, when applicable. see 3.1). Capacitors shall be subjected to 140 percent of the dc rated voltage at the high ambient test temperature for 48 hours +8 hours, -0 hours. During this test, capacitors shall be adequately protected against temporary voltage surges of 10 percent or more of the test voltage. After the test, capacitors shall show no evidence of damage, arcing, or breakdown.



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4.7.3 Radiographic inspection (see 3.7) (ERs only, when applicable. see 3.1). Capacitors shall be tested in accordance with method 209 of MIL-STD-202. The following details shall apply.

a. Radiographic quality: Sufficient definition to determine that specimens are free from defects specified in 3.7.

b. Image-quality indicator: A sample part of the same type as the part being radiographed that contains either an actual or simulated defect which is at least 10 percent smaller than the smallest defect to be detected.

c. Positions of specimen: Two views perpendicular to the terminal axis. After first view, specimen shall be rotated 90 degrees for the second view.

d. Evaluation of images.

(1) Special kind of viewing equipment: Magnifying glass.

(2) Magnification: 10X.

(3) Defects to be sought in specimen: As specified in 3.7.

4.7.4 Thermal shock (see 3.8). Capacitors and retainers shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter: Unless otherwise specified, B, except that during step 3, capacitors shall be conditioned at the high ambient test temperature (see 3.1).

b Measurements before and after cycling: Not applicable.

4.7.5 Seal (see 3.9). Capacitors shall be tested in accordance with method 112, MIL-STD-202. The following details shall apply.

a. Test condition letter: As specified, see 3.1.

b. Examination after test: Capacitors shall be visually examined for evidence of leakage.

4.7.5.1 Seal - alternative test (see 3.9). For capacitors with a liquid impregnant, the following seal test may be substituted: Capacitors shall be placed on a clean sheet of absorbent paper and exposed to the applicable high ambient test temperature for a minimum of 1 hour. The capacitor shall then be visually examined for evidence of leakage of impregnant or filling compounds or bubbles from the seal. Capacitors to be subjected to the salt spray (corrosion), and immersion tests may be excluded from the seal test until after the salt spray (corrosion), and immersion tests are completed.

4.7.6. Dielectric withstanding voltage (see 3.10).

4.7.6.1 Capacitors (see 3.10.1). Capacitors shall be tested in accordance with method 301, MIL-STD-202. The following details and exceptions shall apply:

a. Magnitude of test voltage: As specified in table XV.

b. Nature of potential: dc.

c. Duration of application of test voltage: As specified in table XV.

d. Points of application of test voltage: As specified in table XV.

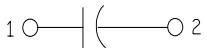
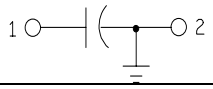
e. Examinations after test: Capacitors shall be visually examined for evidence of breakdown, arcing, or other visible damage.



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4.7.6.2 Sleeving (See 3.10.2) (when applicable. see 3.1). With the capacitor horizontally mounted, a single loop of No. 20 AWG wire shall be secured around the insulating sleeve of the capacitor so that it extends downward from the capacitor 3 inches (76.2 mm) minimum, and the two ends of the wire twisted in about three cross turns. A 1-pound minimum weight shall then be suspended from the looped wire. After exposure in this position for a minimum of 24 hours, at the maximum rated temperature ±3°C, a dc voltage of 4,000 volts minimum, shall be applied between the case and the looped wire The rate of voltage application shall be 500 volts per second and the duration of application of test voltage shall be 15 seconds, minimum.

TABLE XV. Dielectric-withstanding-voltage test details.

| Circuit-diagram symbol | Circuit diagram | Test points | Test connections | Magnitude of test voltage (percent rated dc voltage) | Duration of application of voltage (minutes) |
|---|---|---|---|---|---|
| 1 |  | Terminal to terminal<br>Terminal to case 3/ | 1 to 2<br>1 and 2 to case | 200 1/<br>200 | 1 2/<br>1 |
| 3 |  | Terminal to terminal | 1 to 2 or 1 to case | 200 1/ | 1 2/ |

1/ 175 percent rated dc voltage after immersion, and moisture resistance tests.
2/ For the 100-percent inspection specified in 4.6.1.2, the capacitors shall be subjected, at the option of the manufacturer, to the application of 250 percent of rated dc voltage for not less than 5 seconds, or 200 percent for not less than 15 seconds.
3/ For quality conformance inspection, application of potential may be made between each terminal individually and the case.

4.7.7 Barometric pressure (reduced) (see 3.11) (when applicable. see 3.1), qualification only. Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details and exceptions shall apply.

a. Method of mounting: By normal mounting means.

b. Test condition: Unless otherwise specified (see 3.1), capacitors shall be subjected to a pressure of 0.82 inch of mercury (80,000 feet).

c. Test during subjection to reduced pressure: Unless otherwise specified (see 3.1), a potential equal to that specified in table XVI, as applicable, shall be applied for at least 1 minute between test points specified (see 3.1). A suitable means shall be used to detect momentary or permanent breakdown. Capacitors shall then be visually examined for evidence of damage (see 6.5.5).

TABLE XVI. Barometric-pressure test potentials.

| Case diameter (Inch (mm)) | DC voltage (max) (Volts) 1/ |
|---|---|
| .175 (4.45) | 200 |
| .195 (4.95) | 200 |
| .235 (5.97) | 300 |
| .312 (7.92) | 430 |
| .400 (10.16) | 500 |
| .562 (14.27) and up | 640 |

1/ The dc voltage shall be limited to 200 percent of the dc voltage rating at maximum rated temperature.



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4.7.8 Insulation resistance (see 3.12). Capacitors shall be tested in accordance with method 302, MIL-STD-202. The following details and exceptions shall apply.

a. Test potential:

(1) Insulating sleeves: When applicable (See 3.1). Test condition B.

(2) Terminal to terminal and terminal to case: A potential equal to the rated dc voltage (see 3.1) or 500 V dc, whichever is less.

b. Special conditions: The time constant of the measurement circuit with the capacitor connected shall not exceed 30 seconds.

c. Points of measurement:

(1) Insulating sleeves: The test potential shall be applied between the case and a piece of metal foil placed around the insulating sleeve; the metal foil shall be of such dimension as to allow at least .125 inch (3.18 mm) surface exposure of the insulating sleeve on each end (see 3.12.1).

(2) Terminal to terminal: Insulation resistance shall be measured between terminals at the maximum rated temperature, ±3°C, and at 25°C ±3°C (see 3.12.2).

(3) Terminals to case: When the case is not a terminal, the measurement shall be made between each terminal and the case at 25°C ±3°C (see 3.12.3).

d. Electrification time: 2 minutes.

4.7.9 Capacitance (See 3.13.). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. The following details shall apply.

a. Test frequency: 1,000 Hertz (Hz) ±100 Hz for capacitors whose nominal capacitance does not exceed 1 µF and whose dc voltage rating does not exceed 3,000 volts. For capacitors not within these limits, measurements shall be made at a frequency of 100 Hz ±6 Hz or corrected thereto.

b. Limit of accuracy: Shall be within ±0.l percent.

4.7.10 Dissipation factor (see 3.14). The dissipation factor shall be measured with an ac voltage not greater than 20 percent of the dc voltage rating, at a frequency of 1,000 ±100 Hz, for capacitors whose nominal capacitance does not exceed 1 µF and whose dc voltage rating does not exceed 3,000 volts. For capacitors not within these limits, measurements shall be made at a frequency of 100 Hz ±6 Hz or corrected thereto. Measurement accuracy shall be one of the following.

a. For dial reading: ±2 percent of dial reading or, .001, whichever is greater.

b. For digital readout: ±.001 percent ±2 percent of reading.

4.7.11 Vibration (see 3.1 and 3.15).

4.7.11.1 Low frequency. Capacitors shall be tested in accordance with method 201 of MIL-STD-202. The following details and exceptions shall apply:

a. Tests and measurements prior to vibration: Not applicable.

b. Method of mounting: Securely fastened by normal mounting means, except that capacitors without mounting retainers in sizes 1.562 inch (39.67 mm) long or .670 inch (17.02 mm) in diameter or larger shall be mounted by a supplemental means other than the wire leads. Wire-Lead capacitors shall be secured .5 inch ±.125 inch (12.7 mm ±3.18 mm) from the case. The extraneous leads beyond the .5 inch (12.7 mm) securing point may be removed or supplemental support may be added.



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c. Duration of vibration: For tubular styles: 4 hours (2 hours in each of 2 mutually perpendicular planes). For rectangular styles: 6 hours (2 hours in each of 3 mutually perpendicular planes).

d. Test and measurements during vibration: During the last 30 minutes of vibration in each direction, a signal of 1 kilohertz (kHz) ±0.2 kHz at a level of 1 volt ±0.5 volt shall be placed across the capacitor and measured with suitable ac detecting equipment to determine open -circuits or short-circuits, or intermittent contacts. The accuracy of the detecting equipment shall be sufficient to detect any interruption with a duration of 0.5 ms or greater.

e. Examination after vibration: Capacitors shall be visually examined for evidence of mechanical damage.

4.7.11.2 High frequency. Capacitors and retainers shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens: Capacitors shall be rigidly mounted by the body to a vibration test apparatus. Wire-lead capacitors shall be secured .5 inch ±.125 inch (12.7 mm ±3.18 mm) from the case. The extraneous leads beyond the .5 inch (12.7 mm) securing point may be removed or supplemental support may be added.

b. Electrical-load conditions: During the test, a potential of 125 percent of the dc rated voltage shall be applied between the terminals of the capacitor.

c. Test condition letter: B, with exception as specified (see 3.1).

d. Measurements: During the last cycle in each direction, a signal of 1 kHz ±0.2 kHz at a level of 1 volt ±.05 volt shall be placed across the capacitor and measured with a suitable ac recording device (a permanent record is not necessary for this test) to determine open- or short-circuits, or intermittent contacts. The accuracy of the detecting equipment shall be sufficient to detect any interruption with a duration of 0.5 ms or greater.

e. Measurements and examination after vibration: After the test, with capacitors still mounted on the vibration jig, the insulating sleeve shall be tested for dielectric withstanding voltage, as specified in 4.7.6, with a potential of 2,000 V dc between the case and bracket. Capacitors and retainers shall be visually examined for evidence of mechanical damage.

4.7.12 Salt spray (corrosion) (see 3.16). Capacitors and retainers shall be tested in accordance with method 101, MIL-STD-202. The following details and exceptions shall apply.

a. Applicable salt solution: The salt solution concentration shall be 5 percent.

b. Test condition letter: B (48 hours).

c. Examination after exposure: Capacitors and retainers shall be visually examined for evidence of harmful corrosion, unwrapping of or mechanical damage to insulating sleeves (when applicable), and obliteration of marking.

4.7.13 Immersion (see 3.17). Capacitors and retainers shall be tested in accordance with method 104 of MIL-STD-202. The following details shall apply:

a. Test condition letter B.

b. Measurements after final cycle: For capacitors with insulating sleeves, dielectric withstanding voltage, and insulation resistance at 25°C shall be measured as specified in 4.7.6 and 4.7.8, respectively. The test potential shall be applied between the case and a piece of metal foil placed around the insulating sleeve; the metal foil shall be of such dimension as to allow at least .125 inch (3.18 mm) surface exposure of the insulating sleeve on each end. In addition, dielectric withstanding voltage and insulation resistance at 25°C of all capacitors shall be measured as specified in 4.7.6 and 4.7.8, respectively. Capacitors and retainers shall then be examined for harmful or extensive corrosion and obliteration of marking.

c. Examination after immersion: Capacitors shall be inspected for evidence of extensive corrosion and unwrapping of, or mechanical damage to the insulating sleeves, when applicable.



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4.7.14 Solderability (wire leads only, see 3.1) (see 3.18). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

a. Number of terminations to be tested: Both leads of the capacitor shall be subjected to the solderability test.

b. Depth of immersion in flux and solder: The leads shall be immersed to within .125 inch (3.18 mm) of the capacitor body.

4.7.15 Shock (specified pulse) (see 3.19). Capacitors and retainers shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

a. Mounting: The body of the capacitor shall be securely fastened by mounting retainers. The leads shall be soldered to rigidly supported terminals so spaced that the length of each lead from the capacitor shall be .5 inch ±.125 inch (12.7 mm ± 3.18 mm) from the edge of the supporting terminal.

b. Test condition letter: 1 (100 g's).

c. Electrical loading during shock: During the test, a potential of 125 percent of the dc voltage rating shall be applied between the terminals of the capacitor.

d. Measurements during shock: During the test, observations shall be made to determine intermittent contact or arcing or open-circuiting or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption with a duration of 0.5 ms or greater.

e. Examinations after test: Capacitors shall be visually examined for evidence of breakdown, arcing, fractures, or any other visible mechanical damage. Retainers shall be visually examined for mechanical damage.

4.7.16 Terminal strength (see 3.20). Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter: As specified (see 3.1).

b. Examination after test: The capacitors and terminals shall be examined for mechanical damage.

4.7.17 Dielectric absorption (see 3.21), (when specified. see 3.1). The capacitor shall be charged at the dc voltage rating for 1 hour ±1 minute. The initial surge current shall not exceed 50 milliamperes. At the end of this period, the capacitor shall be disconnected from the power source and discharged through a 5 ohm ±5 percent resistor for 10 seconds ±1 second. The discharge resistor shall be disconnected from the capacitor at the end of the 10 second discharge period, and the voltage remaining on the capacitor (recovery voltage) shall be measured with an electrometer or other suitable device having an input resistance of 10,000 megohms or greater. Recovery voltage shall be read at the maximum voltage within a 15 minute period. The dielectric absorption shall be computed from the following formula.

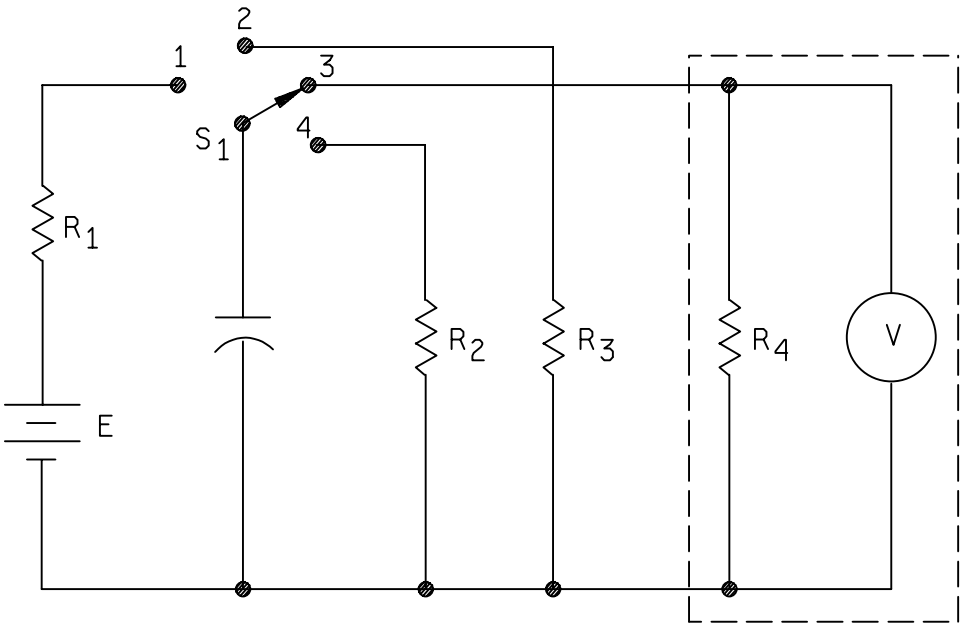
$$d = \frac{V_1 \times 100}{V_2}$$

Where: d = Percent dielectric absorption.
$V_1$ = Maximum recovery voltage.
$V_2$ = Charging voltage.

For an alternate production test method see figure 1.



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E = dc rated dc voltage or 100 volts dc, whichever is less (±2 percent).
$R_1$ and $R_2$ = 1,000 ohms ±20 percent. (This is not critical.)
$R_3$ = 5 ohms ±1 percent.
$R_4$ = 10,000 megohms, minimum.

Suggest use of ±1230A GR electrometer or equivalent set to 10,000 megohms range input resistance.

NOTES:
1. Charge for 5 minutes ±10 seconds with switch in position 1.
2. Switch in position 2 for 5 seconds ±0.5 second.
3. Switch in position 3 for 1 minute.
4. After 1 minute read recovery voltage and compute as a percentage of charge voltage.
5. Switch to position 4, discharge and remove.

FIGURE 1. Typical production dielectric absorption test method.

4.7.18 Stability at low and high temperatures (see 3.22). Capacitors shall be placed in a chamber maintained at -65°C +0°C, -5°C or -55°C +0°C, -5°C (as applicable, see 3.1), and a potential equal to the dc rated voltage shall be applied at this condition for 48 hours ±4 hours. The air within the conditioning chamber shall be circulated. Before capacitors are moved from the conditioning chamber, capacitance shall be measured at the applicable low temperature (see 3.1) as specified in 4.7.9. Capacitance shall then be measured at the following temperature as specified in 4.7.9. (The measurement at each temperature shall be recorded when two successive readings taken at 5-minute intervals indicate no change in capacitance.)

25°C ±5°C
High ambient test temperature (see 3.1)
25°C ±5°C

After the test, capacitors shall be visually examined for evidence of breakdown, arcing, open-circuiting and short-circuiting, and other visible mechanical damage.



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4.7.19 Temperature coefficient (see 3.23) (characteristic P only, see 3.1). Capacitance measurements shall be made after the capacitors have been stabilized at each of the following temperatures: -65°C, +25°C, and the high ambient test temperature (see 3.1). Stability shall be indicated when no change exists in two capacitance measurements made at 5-minute intervals. Temperature coefficient shall be computed from the following formula:

$$T_C = \frac{(C_2 - C_1)\, 10^6}{(T_2 - T_1)\, C_1}$$

Where: $T_C$ = temperature coefficient in ppm/°C.
$C_1$ = capacitance at 25°C in pF.
$C_2$ = capacitance at test temperature in pF.
$T_1$ = 25°C.
$T_2$ = test temperature in degrees C.

TABLE XVII. Life test in hours

| NON-ER capacitors | | | |
|---|---|---|---|
| Characteristic | Qualification | Group B | |
| All characteristics except M<br>M | 240 +48, -0 1/<br>2,000 +72, -0 | 240 +48, -0 1/<br>2,000 +72, -0 | |
| **ER capacitors** | | | |
| Style | Qualification | Group B | Extended life |
| CQR39, 42, and 43<br>FRLs M, P, R, S, for all styles except CQR39, 42, and 43 | 240 +48, -0<br>2,000 +72, -0 | 240 +48, -0<br>2,000 +72, -0 | ---<br>4,000 +72, -0 |

1/ Test time for characteristic K tubular capacitors with voltage rating of 1,000 volts shall be 500 ± 12 hours.

4.7.20 Life (see 3.24).

4.7.20.1 Life test (qualification inspection). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply.

a. Distance of temperature measurements from specimens, inches: Not applicable.

b. Test temperature and tolerance: At the applicable high-test temperature, +4°C, -0°C (see 3.1).

c. Operating conditions: Capacitors shall be subjected to 140 percent of the rated dc voltage. The surge current shall be limited to between 5 milliamperes and 1 ampere. When necessary, a suitable current-limiting resistor shall be used.

d. Test duration: See table XVII.

e. Measurements during and after exposure: At the conclusion of the test, the capacitors shall be returned to the inspection conditions specified in 4.3 and shall be visually examined for evidence of mechanical damage and obliteration of marking; insulation resistance at +25°C, capacitance, and dissipation factor shall be measured as specified in 4.7.8, 4.7.9, and 4.7.10, respectively.

4.7.20.2 Life test (group B inspection) (see 3.24). Except as specified in the following, capacitors shall be tested as specified in 4.7.20.1.

a. Test duration: See table XVII.

b. Measurements during and after exposure: Measurements, as specified in 4.7.20.1e, shall be taken at the following intervals of exposure: During the first hour; 240 hours +48 hours, -0 hours; and where applicable, 1,000 hours +48 hours, -0 hours; and 2,000 hours +72 hours, -0 hours.



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4.7.20.3 Extended life (see 3.24). Sample units that have been subjected to 2,000 hours of life test as specified in 4.7.20.1 and 4.7.20.2, shall remain on test for an additional 4,000 hours +72 hours. -0 hours. After acceleration of the 6,000-hour life test time capacitors shall be measured as specified in 4.7.20.1e.

4.7.21 Fungus (see 3.25). Capacitors shall be tested in accordance with method 508 of MIL-STD-810. Pretest and post-test measurements are not required.

4.7.22 Resistance to solvents (see 3.26) Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

a. Portion of specimen to be brushed: That portion on which marking is present.

b. Number of specimens to be tested: As specified in applicable inspection tables.

c. Permissible extent of damage: As specified in 3.26.

4.7.23 Resistance to soldering heat (see 3.27) (applicable to wire-lead capacitors only). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

a. Test condition letter: C

b. Cooling time prior to measurement after test: 10 minutes ±1 minute.

c. Measurements after test: Insulation resistance, capacitance, and dissipation factor shall be measured at 25°C ±3°C as specified in 4.7.8, 4.7.9, and 4.7.10, respectively.

4.7.24 Moisture resistance (see 3.28). Capacitors and retainers shall be tested in accordance with method 107, test condition A, MIL-STD-202, except that no measurement shall be made before and after cycling. Capacitors shall then be tested in accordance with method 106, MIL-STD-202. The following details and exceptions shall apply:

a. Initial measurements: Not applicable.

b. Polarization voltage: During steps 1 through 6, inclusive, a dc potential of 100 volts shall be applied across the terminals of 50 percent of the capacitors. No potential shall be applied to the remaining 50 percent of the capacitors.

c. Loading voltage: Not applicable.

d. Final measurements: After the final cycle, the capacitors shall be conditioned at 25°C ±5°C and a relative humidity of 50 percent ±5 percent for a period of at least 22 hours but not more. Dielectric withstanding voltage and insulation resistance at 25°C shall be measured on insulating sleeves as specified in 4.7.6 and 4.7.8, respectively. The test potential shall be applied between the case and a piece of metal foil placed around the insulating sleeve. The metal foil shall be of such dimension as to allow at least .125 inch (3.18 mm) surface exposure of the insulating sleeve on each end. In addition, dielectric withstanding voltage, insulation resistance, capacitance, and dissipation factor at 25°C, of all capacitors shall be measured as specified in 4.7.6, 4.7.8, 4.7.9, and 4.7.10, respectively.

After the test, capacitors and retainers shall be visually examined for evidence of harmful corrosion and obliteration of marking; capacitors shall be examined for unwrapping of or mechanical damage to insulated sleeves, when applicable.

4.7.25 Flashpoint of impregnant or filling compound (see 3.29). The flashpoint of impregnant or filling compound shall be measured as specified in ASTM D92, except that the fire point and precision do not apply. "Impregnant or filling compound" shall be substituted for the word "oil" throughout the test method.



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5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Capacitors covered by this specification are intended for use in electronic equipment. Capacitors covered by this specification are unique due to the fact that these devices must be able to operate satisfactorily in military systems under the following demanding conditions: 15 G's of high and low frequency vibration, 100 G's of shock (specified pulse), operate in extreme temperatures (-65°C to +170°C), and operate in moist environments. These capacitors also offer high reliability that is verified under a qualification system. Commercial components are not designed to withstand these military environmental conditions.

6.1.1 Polystyrene dielectric capacitors (characteristic P). Capacitors of polystyrene dielectric, because of their low dielectric absorption and radiofrequency losses, are intended primarily for use in calculators, computers, integrators, time-base oscillators, laboratory standards, and other pulse applications. The outstanding characteristics of these capacitors are low temperature coefficient and stability.

6.1.2 Polyethylene terephthalate dielectric capacitors (characteristic M). Capacitors of polyethylene terephthalate dielectric are intended for use in high temperature applications similar to those served by hermetically sealed paper capacitors, but where higher insulation resistance at the upper temperature limits is required.

6.1.3 Paper and polyethylene terephthalate dielectric capacitors (characteristics E, F, G, and K). Capacitors of paper and polyethylene terephthalate dielectric are intended for applications where small case sizes and high temperature operation are required.

6.1.4 Polytetrafluoroethylene dielectric capacitors (characteristic T). Capacitors of polytetrafluoroethylene dielectric are intended for high temperature applications where high insulation resistance, small capacitance change, and low dielectric absorption are required. These capacitors exhibit excellent insulation resistance values at high temperatures.

6.1.5 Polycarbonate dielectric capacitors (characteristic Q). Capacitors of polycarbonate dielectric are especially suitable for use in tuned circuits and precision timing due to their capacitance stability and minimum capacitance change with temperature.

* 6.1.6 Polyphenylene sulfide foil dielectric capacitors (characteristic V). Capacitors of polyphenylene sulfide foil dielectric are intended as alternatives for polycarbonate dielectric capacitors (characteristic Q). Characteristic Q dielectric capacitors are being inactivated for new design since polycarbonate material is no longer made.

6.2 Acquisition data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet and the complete PIN (see 1.2.1 and 3.1).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

d. Whether removable mounting retainers are to be supplied with capacitors (see 3.1).

e. If remarking of parts is required to indicate FRL (see 3.30.5).



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f. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Liquid-impregnanted capacitor. A liquid-impregnanted capacitor is a capacitor in which a liquid impregnant is dominantly contained within the capacitor element, but does not occupy substantially all of the case volume not required by the capacitor element and its connection.

6.3.2 Liquid-filled capacitor. A liquid-filled capacitor is a capacitor in which a liquid impregnant occupies substantially all of the case volume not required by the capacitor element and its connections. Space may be allowed for the expansion of the liquid with temperature variations.

6.3.3 Nonmagnetic case capacitor. A nonmagnetic case capacitor is a capacitor in which no part of the case or retainer is made of magnetizable material; however, the end seal may be made of a material having a ferrous metal content in order to effect a glass-to-metal seal.

6.3.4 Hermetic seal. For the purpose of this specification, a hermetically sealed capacitor is one in which the capacitive element is contained within a sealed enclosure of ceramic, glass or metal, or combinations thereof, where sealing is accomplished by material fusion, welding, brazing, or soldering. The capacitor shall be capable of passing the seal test specified in 4.7.5 or of meeting a leak-rate requirement of not more than 1.76 by $10^{6}$ cubic centimeters ($cm^{3}$) per second, when determined by any other method having sensitivity equal to or better than the stated limit. When capacitors are tested as specified in 4.7.5, there shall be no continuous visible stream of bubbles.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the QPL whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the QPL is the Defense Supply Center, Columbus (DSCC-VQP), 3990 East Broad Street, Columbus, Ohio 43216-5000. Application for qualification tests shall be made in accordance with SD-6, “Provisions Governing Qualification”, copies of which may be obtained on application to Defense Automation and Production Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.5 Application information.

6.5.1 Mounting. Capacitors covered by this specification shall be mounted by a retainer or clasp or should be potted when vibration or shock are likely to be encountered in service.

6.5.2 Voltage derating with temperature (for characteristic M). Characteristic M tubular capacitors may be used in applications up to 125°C with voltage derating as indicated in figure 2. However, insulation resistance requirements at 125°C cannot be expected to exceed 100 megohm-microfarads; greater capacitance change may be encountered, and life expectancy of the unit will be reduced.

6.5.3 Energy content (in watt-seconds). The energy content of a fully charged dc capacitor is determined by the following formula:

$$W = \frac{CE^2}{2}$$

Where: W = energy content in watt-seconds.
C = nominal capacitance in farads.
E = dc voltage rating in volts.

6.5.4 AC component. The rating given is the steady-state dc voltage, or the sum of the dc voltage and the peak ac voltage, provided that the peak ac voltage does not exceed 20 percent of the rating at 60 Hz, 15 percent at 120 Hz, or 1 percent at 10,000 Hz. Where heavy transient or pulse currents are encountered, the requirements of this specification are not sufficient to guarantee satisfactory performance, and due allowance must therefore be made in the selection of a capacitor.



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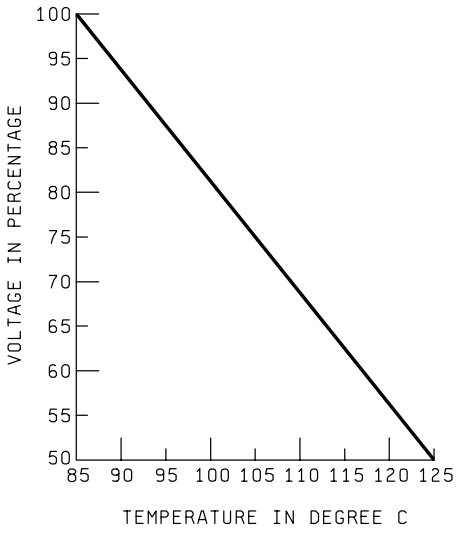



FIGURE 2. Intended operating temperature.

6.5.5 Barometric pressure (reduced) (see 4.7.7).

6.5.5.1 Ceramic-cased or glass-cased tubular capacitors. The dc voltage that may be applied to ceramic-cased or glass-cased tubular capacitors at altitudes from 50,000 to 100,000 feet may be obtained from figure 3.

6.5.5.2 Metal-cased tubular capacitors. The dc voltage that may be applied to metal-cased tubular capacitors at altitudes other than 80,000 feet may be obtained from figure 4, except that the dc voltage rating must not be exceeded.

6.5.6 Representation of minimum insulation resistance. A graphical representation of minimum insulation resistance, as specified in the applicable specification sheet, is shown in figure 5.

6.5.7 Life at temperatures and voltages below and above rated. The failure rates used in this specification are referred to operation at rated voltage at the maximum rated temperature. The sampling plans and failure-rate determinations throughout the specification assume an acceleration factor of 5 for the life test conducted at the maximum rated temperature and 140 percent of rated voltage. Lower failure rates than those for which the manufacturer has obtained qualification may be achieved by operating the capacitors at lower voltage, or at lower temperatures, or both. Factors by which failure rates are to be multiplied under conditions other than maximum conditions are shown on figure 6 (characteristic K only).



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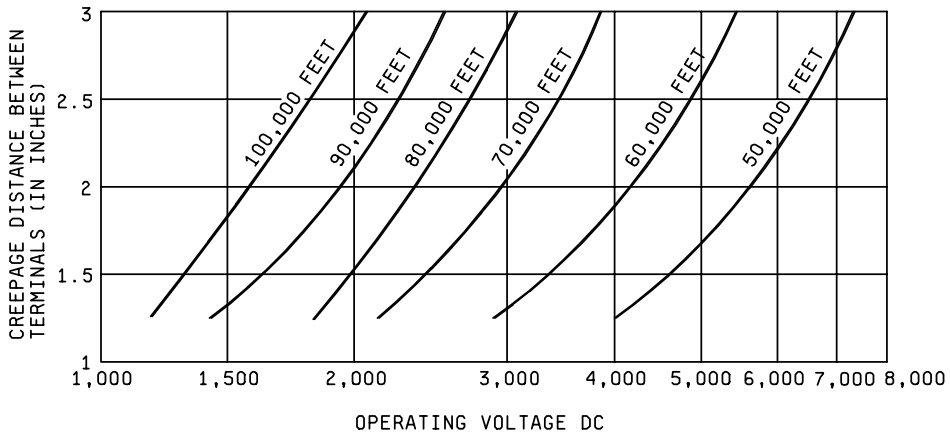



FIGURE 3. Permissible operating voltage at altitudes from 50,000 to 100,000 feet.

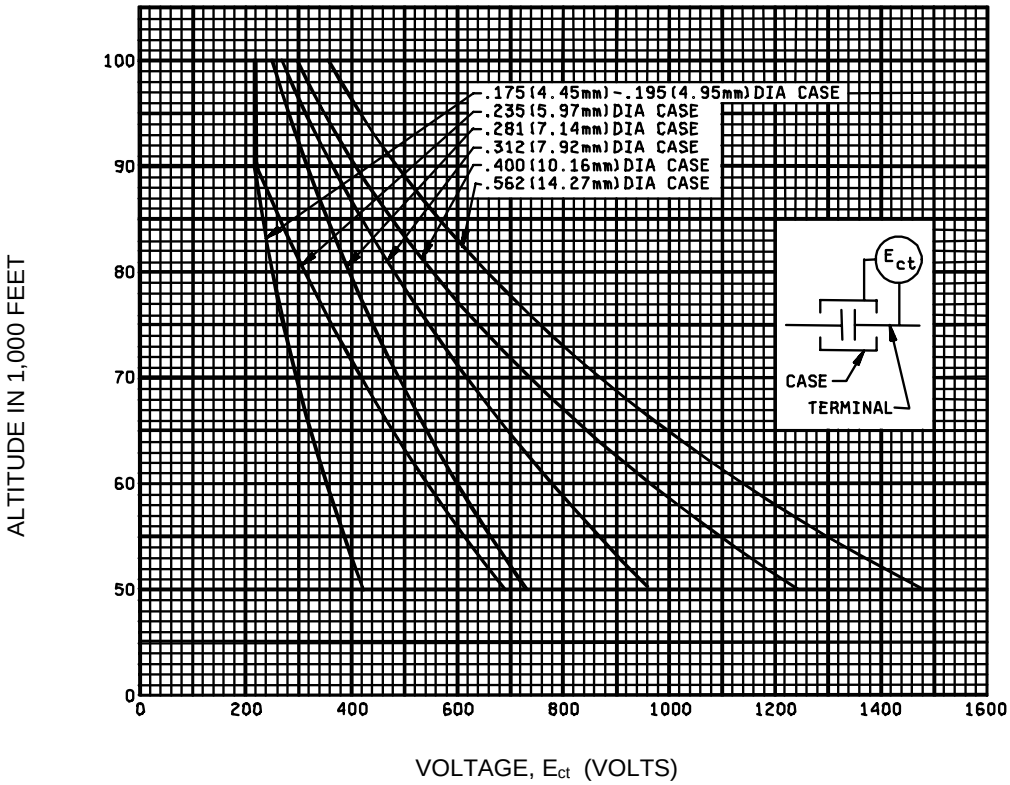



FIGURE 4. Permissible operating voltage at altitudes other than 80,000 feet (metal-cased tubular).



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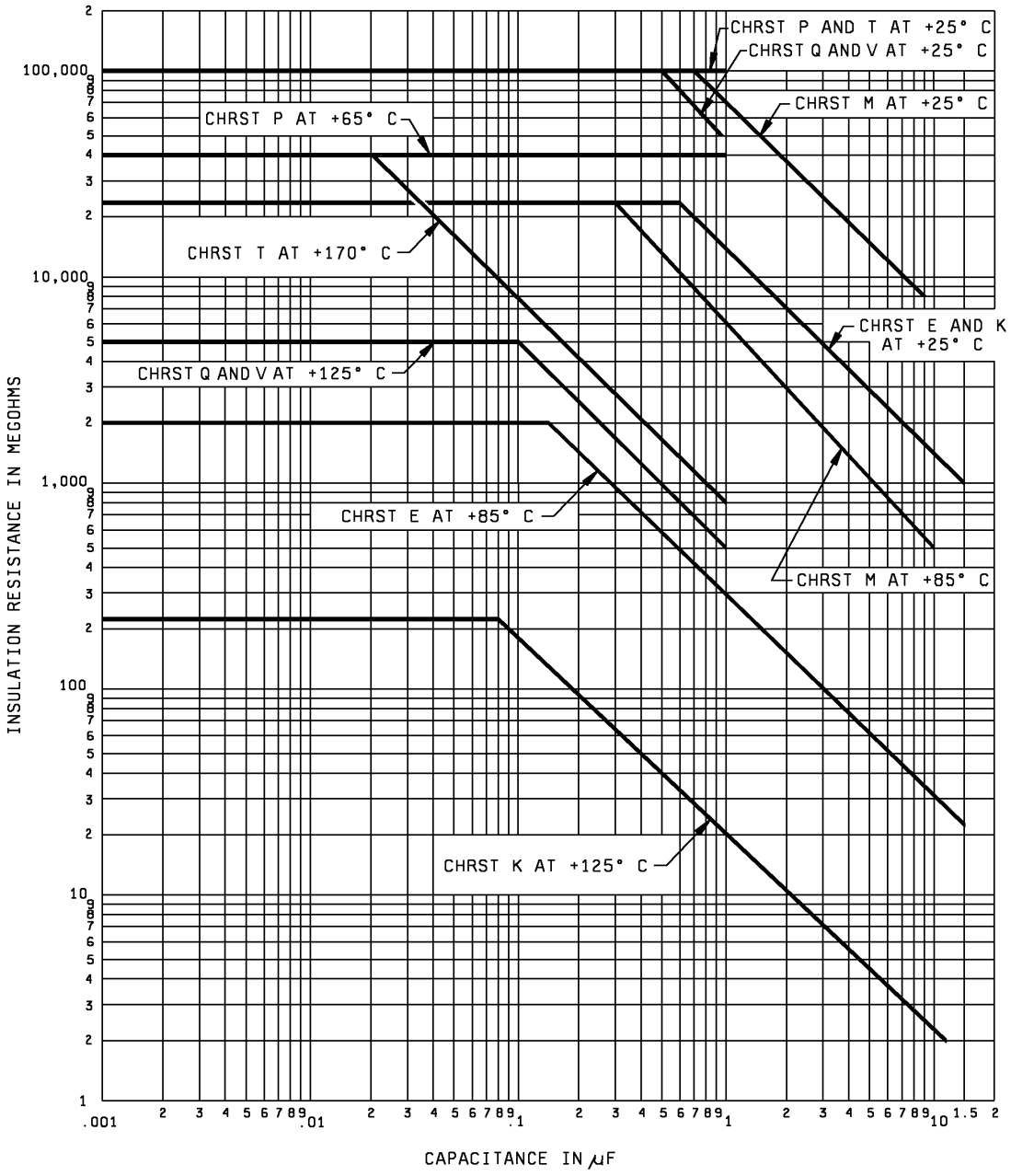
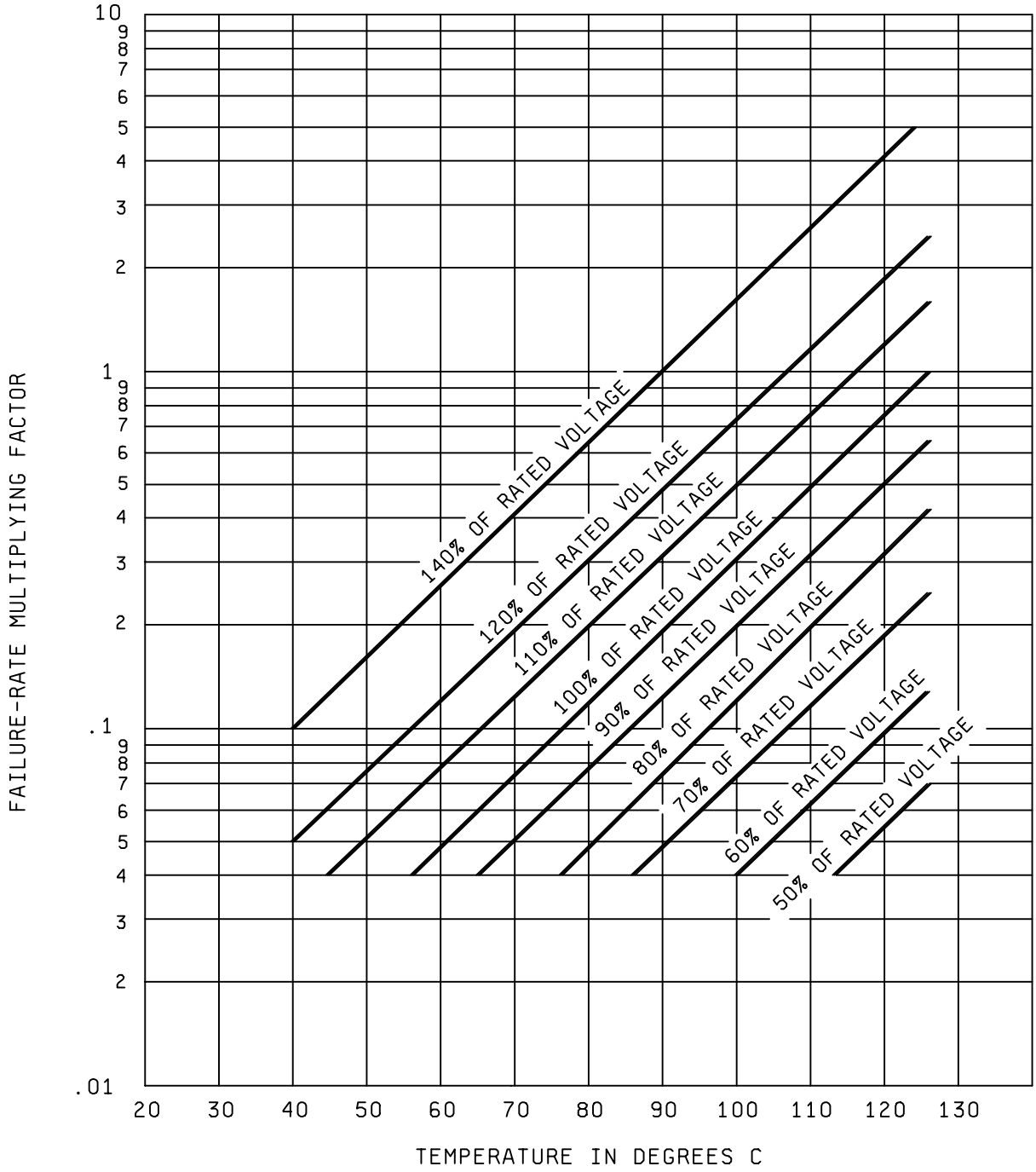



FIGURE 5. Graphical representation of minimum insulation resistance by dielectric as specified in the applicable specification sheet.



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NOTES: Curves for 100 percent of rated voltage and higher are based on the 5th power rule and curves for 90 percent of rated voltage and below are based on the 4th power rule.

FIGURE 6. Life at temperature and voltages relative to percent rating (characteristic K only).



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6.6 Failure rate L. Failure rate L has been canceled from this specification.

6.7 Tin plated finishes. Tin plating is prohibited (see 3.5.2.5) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter, refer to ASTM 8545 (Standard Specification for Electrodeposited Coating of Tin.)

6.8 PIN. This specification requires a PIN that describes codification and/or classification and appropriate references to associated documents (see 1.2 and 3.1).

* 6.9 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Table XIX lists the Environmental Protection Agency (EPA) top seventeen hazardous materials targeted for major usage reduction. If any of these hazardous materials are required, it is recommended that it be used only when other materials cannot meet performance requirements.

TABLE XIX. EPA top seventeen hazardous materials.

| | | |
|---|---|---|
| Benzene | Dichloromethane | Tetrachloroethylene |
| Cadmium and compounds | Lead and compounds | Toluene |
| Carbon tetrachoride | Mercury and compounds | 1,1,1 - Trichloroethane |
| Chloroform | Methyl ethyl ketone | Trichloroethyene |
| Chromium and compounds | Methyl isobutyl ketone | Xylenes |
| Cyanide and compounds | Nickel and compounds | |

6.10 Subject term (key word) listing.

Part Per Million (ppm)
Statistical process control (SPC)

6.11 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

* 10.1 Scope. This appendix details the procedure for submission of samples for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance only.

20. APPLICABLE DOCUMENTS

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1276 - Leads for Electronic Component Part.

30. SUBMISSION

30.1 Single-type capacitor submission. The following details apply.

a. Sample size: As specified in table IX.

b. Sampling criteria: Sample units of the same style, terminal, circuit, characteristic, rated voltage, capacitance, capacitance tolerance, and vibration grade will be submitted.

c. Extent of qualification:

(1) Circuit diagram 1 will qualify circuit diagram 3 in the same capacitor design.

(2) Qualification in one characteristic does not constitute qualification in any other characteristic.

(3) Any capacitance tolerance may represent all other authorized capacitance tolerances.

(4) Vibration grade 3 will qualify vibration grade 1 provided that the designs of the capacitors are the same.

(5) Qualification of insulated styles will be the basis for qualification of uninsulated styles of the same design.

(6) Styles CQ09, CQ13, CQR09, CQR13, CQR19, CQR29, CQR33, CQR39, and CQR49 require individual voltage and characteristic qualification. Style CQ09 will qualify styles CQ08, and CQ12; CQR09 will qualify style CQR12; CQR29 will qualify CQR32, and CQR39 will qualify CQR42, provided the type submitted has the same impregnant or filling compound. The highest capacitance value submitted in each voltage rating will confer qualification of all units of the same style and lesser capacitance with the same voltage rating.

Style CQ20 qualification is limited to the watt-second rating qualified and lesser values in the same voltage rating.

Qualification of ER styles will qualify applicable non-ER styles of the same characteristic, voltage rating, and case size.



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APPENDIX

Qualification of style CQ72 is restricted to those capacitance values and voltage ratings equal to and less than the ratings represented within the same watt-second group of the submission. Except type CQ72D1ES105K3 will qualify units in the same watt-second group (characteristic E) with 12,500 volt ratings provided that type CQ72D1ET105K3 qualification is complete; otherwise, type CQ72D1ET504K3 will qualify lesser watt-second ratings of 12,500 volt units in characteristic E. Type CQ72D1KT254K3 will qualify type CQ72D1KS504K3 rated at 25 watt-seconds. Terminal style E4 will represent E3.

Qualification approval granted for one of the lower failure rate levels will include qualification approval for all of the higher failure rate levels as indicated in table XX.

TABLE XX. FR qualification data.

| Failure rate level | Will qualify failure rate level |
|---|---|
| S | S, R, P, M |
| R | R, P, M |
| P | P, M |
| M | M |

* 30.2 Combined type submission. Qualification of the types submitted will be a basis for qualification of the style represented, based on the same provisions stated in 30.1 with the following exceptions: Voltage ratings are grouped in table XXI for the styles shown; samples are divided equally between the two voltage ratings of each group. Voltage ratings not listed In the table are qualified separately. Styles are grouped as shown in table XXII.

* TABLE XXI. Voltage rating groups.

| Styles CQ09, CQ13, CQR09, CQR13, CQR19, CQR29, CQR33, CQR39, CQR43 | Style CQ20 |
|---|---|
| Voltage rating | Voltage rating |
| 600 and 400 | 15000 and 12500 |
| 200 and 100 | 10000 and 7500 |
| 50 and 30 | 5000 and 3000 |
| | 2000 and 1000 |

NOTE: In view of the relatively few sample units of each type represented, a combined-type submission will be treated similarly to a single-type submission, and the failure of one type will disqualify the entire submission.



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APPENDIX

TABLE XXII. Examples of combined-type submissions. 1/

| Styles | Types to be submitted | Number of sample units to be submitted | DC voltage rating | Watt-second rating |
|---|---|---|---|---|
| CQ08, CQ09, CQ12, and CQ13 | CQ09A1KC105J3 | 23 | 200 | .0200 |
| | CQ13A1KC105J3 | 23 | 200 | .0200 |
| | CQ09A1KE474J3 | 23 | 400 | .0376 |
| | CQ13A1KF474J3 | 23 | 600 | .0846 |
| | CQ09A1KG474J3 | 23 | 1,000 | .235 |
| | CQ13A1KG474J3 | 23 | 1,000 | .235 |
| | CQ09A1MZ106J3 | 23 | 30 | .0045 |
| | CQ13A1MA125J3 | 23 | 50 | .0015 |
| | CQ09A1MB685J3 | 23 | 100 | .0340 |
| | CQ13A1MC105J3 | 23 | 200 | .0200 |
| | CQ09A1ME105J3 | 23 | 400 | .0800 |
| | CQ13A1MF684J3 | 23 | 600 | .1224 |
| | CQ09A1MG224J3 | 23 | 1,000 | .110 |
| | CQ13A1MG224J3 | 23 | 1,000 | .110 |
| | CQ09A1PA105J3 | 23 | 50 | .00125 |
| | CQ13A1PA105J3 | 23 | 50 | .00125 |
| | CQ09A1PC474J3 | 23 | 200 | .0094 |
| | CQ13A1PC474J3 | 23 | 200 | .0094 |
| | CQ09A1PE154J3 | 23 | 400 | .0120 |
| | CQ13A1PF224J3 | 23 | 600 | .0396 |
| | CQ09A1TC104J3 | 23 | 200 | .0020 |
| | CQ13A1TC104J3 | 23 | 200 | .0020 |
| | CQ09A1TE334J3 | 23 | 400 | .0264 |
| | CQ13A1TF154J3 | 23 | 600 | .0275 |
| CQ20 | CQ20A1KG105K1 | 23 | 1,000 | .500 |
| | CQ20C1KJ684K1 | 23 | 2,000 | 1.36 |
| | CQ20A1KL334K1 | 23 | 3,000 | 1.485 |
| | CQ20C1KN154K1 | 23 | 5,000 | 2.375 |
| | CQ20-1KR683K1 | 23 | 7,500 | 1.862 |
| | CQ20-1KS393K1 | 23 | 10,000 | 1.950 |
| | CQ20-1KT223K1 | 23 | 12,500 | 1.714 |
| | CQ20-1KU153K1 | 23 | 15,000 | 1.688 |
| CQ72 | CQ72B1EG105K3 | 23 | 1,000 | .5 |
| | CQ72E1KG105K3 | 23 | 1,000 | .5 |
| | CQ72E1EN104K3 | 23 | 5,000 | 1.25 |
| | CQ72D1ES104K3 | 23 | 10,000 | 5.0 |
| | CQ72D1KS104K3 | 23 | 10,000 | 5.0 |
| | CQ72D1ES105K3 | 23 | 10,000 | 50.0 |
| | CQ72D1KT154K3 | 23 | 12,500 | 19.531 |
| | CQ72D1ET105K3 | 23 | 12,500 | 78.125 |
| CQR07 | CQR07A1QA105J3M | 82 | 50 | .00125 |
| | CQR07A1QB105J3M | 41 | 100 | .005 |
| | CQR07A1QC684J3M | 41 | 200 | .0136 |
| | CQR07A1QE474J3M | 41 | 400 | .0376 |
| | CQR07A1QF224J3M | 41 | 600 | .0432 |

See footnote at end of table.



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TABLE XXII. Examples of combined-type submissions - Continued. 1/

| Styles | Types to be submitted | Number of sample units to be submitted | DC voltage rating | Watt-second rating |
|---|---|---|---|---|
| CQR09, CQR12, and CQR13 | CQR09A1KC105J3M | 41 | 200 | .020 |
| | CQR13A1KC105J3M | 41 | 200 | .020 |
| | CQR09A1KE474J3M | 41 | 400 | .0376 |
| | CQR13A1KF474J3M | 41 | 600 | .0846 |
| | CQR09A1KG474J3M | 41 | 1,000 | .235 |
| | CQR13A1KG474J3M | 41 | 1,000 | .235 |
| CQR19 | CQR19A1KD105J3M | 41 | 300 | .045 |
| | CQR19A1KE105J3M | 41 | 400 | .080 |
| | CQR19A1KF474J3M | 82 | 600 | .0846 |
| CQR29, CQR32, and CQR33 | CQR291MZ106J3M | 41 | 30 | .0045 |
| | CQR331MA125J3M | 41 | 50 | .0015 |
| | CQR291MB685J3M | 41 | 100 | .0340 |
| | CQR331MC105J3M | 41 | 200 | .0200 |
| | CQR291ME105J3M | 41 | 400 | .0800 |
| | CQR331MF684J3M | 41 | 600 | .1224 |
| | CQR291MG224J3M | 41 | 1,000 | .1100 |
| | CQR331MG224J3M | 41 | 1,000 | .1100 |
| CQR39, CQR42, and CQR43 | CQR391KB105K3M | 41 | 100 | .0050 |
| | CQR431KC105K3M | 41 | 200 | .0200 |
| | CQR391KE474K3M | 41 | 400 | .0376 |
| | CQR431KF474K3M | 41 | 600 | .0846 |
| | CQR391KG474K3M | 41 | 1,000 | .2350 |
| | CQR431KG474K3M | 41 | 1,000 | .2350 |
| CQ05 and CQ10 | CQ05A1KE105K3 | 23 | 400 | .08 |
| | CQ10A1KE105K3 | 23 | 400 | .08 |
| | CQ05A1KG474K3 | 23 | 1,000 | .235 |
| | CQ10A1KG474K3 | 23 | 1,000 | .235 |
| CQ11 | CQ11A1KE474K3 | 23 | 400 | .04 |
| | CQ11A1KF684K3 | 23 | 600 | .122 |
| CQR44 | CQR44A1VA105J3M | 82 | 50 | .00125 |
| | CQR44A1VB105J3M | 41 | 100 | .005 |
| | CQR44A1VC684J3M | 41 | 200 | .0136 |
| | CQR44A1VE474J3M | 41 | 400 | .0376 |
| | CQR44A1VF224J3M | 41 | 600 | .0432 |

1/ This table is set up as a guide; for other combinations, reference is made to the text.

30.3 Impregnant of filling compound. A minimum of .25 pound of each impregnant or filling compound used in the sample units for which qualification is sought shall be submitted.

40. SOLDER DIP (RETINNING) LEADS

40.1 Solder dip (retinning) leads. The manufacturer (or authorized category C distributor) may solder dip/retin the leads of product supplied to this specification provided the solder dip process or an equivalent process has been approved by the qualifying activity.



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40.2 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options.

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276, the manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the product. (NOTE: The 200 microinch maximum thickness requirement is not applicable.)

b. When the lead originally qualified was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in 4.2a, approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Thirty samples of any capacitance value for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder process, the capacitors shall be subjected to the electrical tests of group A inspection. No defects are allowed.

(2) Ten of the 30 samples shall then be subjected to the solderability tests. No defects are allowed.

(3) The remaining 20 samples shall be subjected to the resistance to soldering heat test followed by the moisture resistance test. No defects are allowed.

40.3 Solder dip/retinning options. The manufacturer (or authorized category C distributor) may solder dip/retin as follows:

a. As a corrective action if the lot fails the group A solderability test.

b. After the group A inspection has been completed and, following the solder dip/retinning process, the capacitance, DF, and IR (at 25°C) measurements shall be performed on 100 percent of the lot. The percent defective allowable (PDA) for the electrical measurements shall be as for the subgroup 1 tests. (NOTE: If x-ray and hermetic seal are required in the group A, subgroup 1 tests, these tests shall be repeated.) Following these tests, the manufacturer shall submit the lot to the group A solderability test as specified in 4.7.14.



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - MC
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

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| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER** MIL-PRF-19978J | **2. DOCUMENT DATE** (YYMMDD) 030325 |
|---|---|---|

**3. DOCUMENT TITLE** Capacitors, Fixed, Plastic (or Paper-Plastic) Dielectric, (Hermetically Sealed in Metal, Ceramic or Glass Cases), Established and Non-Established Reliability, General Specification for

**4. NATURE OF CHANGE** (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

**5. REASON FOR RECOMMENDATION**

| **6. SUBMITTER** | | |
|---|---|---|
| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Incl Area Code)<br>(1) Commercial<br>(2) DSN<br>(If applicable) | 7. DATE SUBMITTED<br>(YYYYMMDD) |

| **8. PREPARING ACTIVITY** | |
|---|---|
| a. NAME: Michael Radecki<br>Defense Supply Center, Columbus<br>ATTN: VAT | b. TELEPHONE (Include Area Code)<br>(1) Commercial (614) 692-0561 (2) DSN 850-0561<br><br>(E-Mail) CapacitorFilter@dscc.dla.mil |
| c. ADDRESS (Include Zip Code)<br><br>PO Box 3990<br>Columbus OH 43216-5000 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, Virginia 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 |



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